

Minimizing Autocorrelation via Time-Demeaning the Variables

In prior work, we have “first-differenced” variables (i.e. created year over year change values) for inclusion in regressions using panel data to minimize autocorrelation. However, there is a potentially more efficient econometric approach that involves “time-demeaning” variables (by subtracting the overall mean of each unit or place from the annual value). Time-demeaning does not drop the first year of data like first-differencing and it can sometimes be accomplished with fewer coding steps. This exercise is designed to teach key time-demeaning concepts.

Source File: Download the “crime_predictors_1990_2015.xlsx” file from the “Crime Data” subdirectory on OneDrive.

1. Create a dataframe called “crime” from the downloaded crime_predictors_1990_2015.xlsx dataset. Include only jurisdictions where months_reported contains ‘Dec’ (or December).
2. Create a smaller dataframe called “michigan_crime” that includes only Michigan jurisdictions (i.e. agencies) with over 10,000 population and where total_officers is not missing (!is.na). Keep only the following variables: ori, agency, year, population, violent_crime_rate, total_officers.
3. Create a new dataframe “michigan_avg” that contains the average for three variables, (violent_crime_rate, population, and total_officers) for each agency over all of the years. To calculate the averages, use the group_by and summarize functions.
4. Merge the summarized dataframe (michigan_avg) with michigan_crime to obtain time demeaned variables (population_dm, violent_dm, officer_dm). Create these time demeaned variables by subtracting the mean values from the annual values.
5. Use the cor function to find the correlations between the three original values (population, violent_crime_rate, total_officers). The simplest way to obtain a correlation matrix (table) for more than two variables is to create a new dataframe (new_df) that includes only the variables that are needed via the dplyr “select” command. Then run: cor(new_df)
6. Next, use the cor function to find the correlations between the three demeaned values (population_dm, violent_dm, officers_dm). Notice that the original values will typically have a higher (stronger) correlation than the demean values due to autocorrelation.
7. Lastly, run two linear regression models (one using original values, the other using time demeaned values). In R, the basic syntax for a linear regression is:

```
df_result -> lm(dependent_variable ~ predictor_var1 + predictor_var2, dataframe_name)
summary(df_result)
```

Note: There is a faster way to time-demean variables for use in regression models. See the code example on the next page.

More efficient syntax to time demean the independent and dependent variables for use in panel data regressions:

```
school_dm_final <- school_dm_final1%>%  
group_by(fips_county) %>%  
mutate(across(c(county_enroll:employment_rate), ~ .x - mean(.x), .names = "{col}_DM"))
```